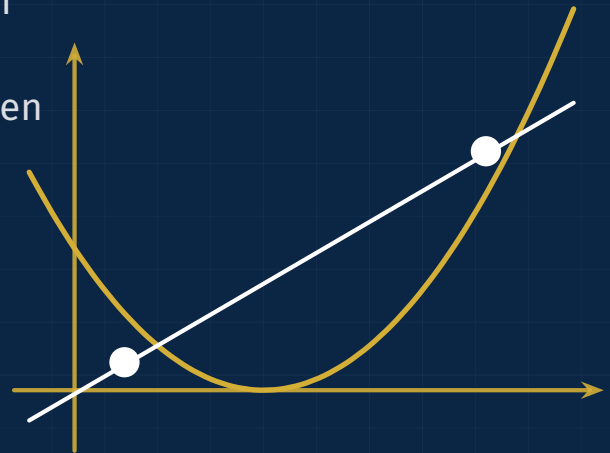


DR ESLAM AHMED
eliteigcse.com

CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 2

Coordinate Geometry

Three tools — midpoint, gradient, length — and one line formula $y - y_1 = m(x - x_1)$. Draw a sketch, label every point, then calculate.

**5**

VARIANTS

40

PAST-PAPER PARTS

28

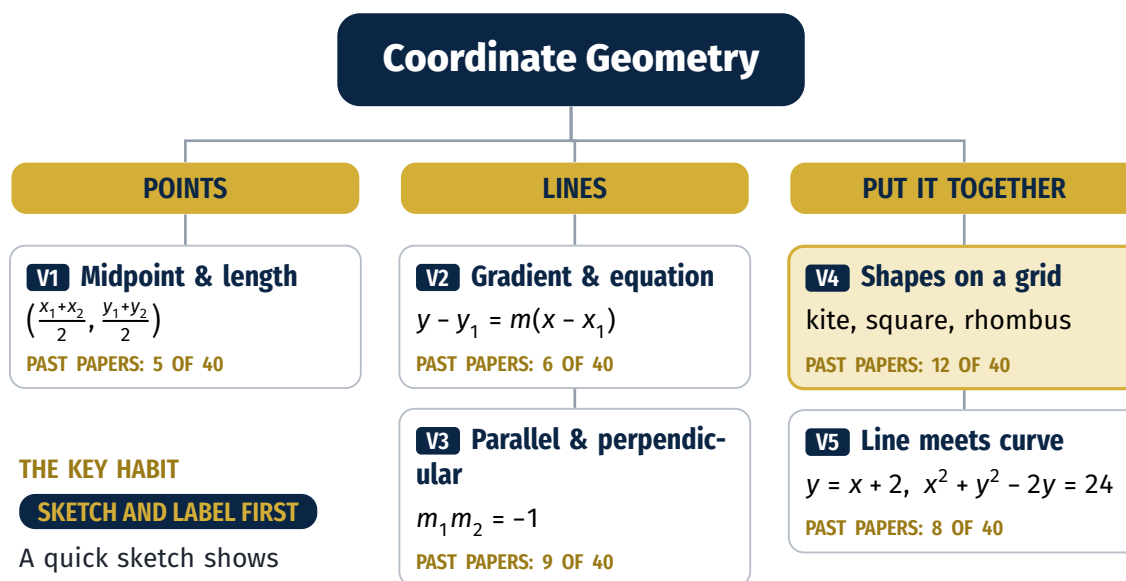
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Midpoint & length**V2** Gradient & equation**V3** Parallel & perpendicular**V4** Shapes on a grid**V5** Line meets curve

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Coordinate geometry at a glance



Your notes


★ How often does each variant come up?

32 past-paper questions (Jan 2020 → Nov 2025) hold 40 parts; 28 practice parts — grouped by type.

	PAST PAPERS (OUT OF 40)	PRACTICE PARTS
V1 Midpoint, length	5	14
V2 Gradient, equation	6	8
V3 Parallel, perpendicular	9	2
V4 Shapes on a grid	12	2
V5 Line meets curve	8	2

WHAT THIS TELLS YOU On every paper, usually twice. The big marks (5–7) are shapes on a grid and a line meeting a curve — both are just V1–V3 joined together, so master the three tools first.

BEFORE YOU START

The big idea



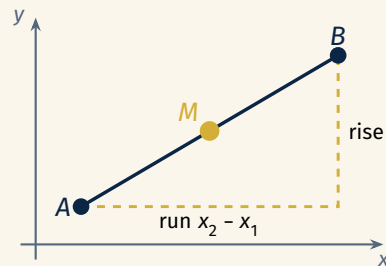
Your notes

Two points give you everything.

From $A(x_1, y_1)$ and $B(x_2, y_2)$: the **midpoint** (average), the **gradient** (rise over run) and the **length** (Pythagoras).

A point and a gradient give the **line**:

$$y - y_1 = m(x - x_1).$$


 The four formulas

Midpoint

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Average the x's and the y's.

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Rise over run.

Length

$$\sqrt{(\Delta x)^2 + (\Delta y)^2}$$

Pythagoras on the run and rise.

Line

$$y - y_1 = m(x - x_1)$$

One point, one gradient.

 Gradients of related lines

Parallel

$$m_2 = m_1$$

Same gradient.

Perpendicular

$$m_2 = -\frac{1}{m_1}$$

Flip and change sign:

$$2 \rightarrow -\frac{1}{2}.$$

Read m

$$2y + 7x = 10 \Rightarrow y =$$

$$-\frac{7}{2}x + 5$$

Make y the subject first.



CALCULATOR FIRST, THEN CHECK

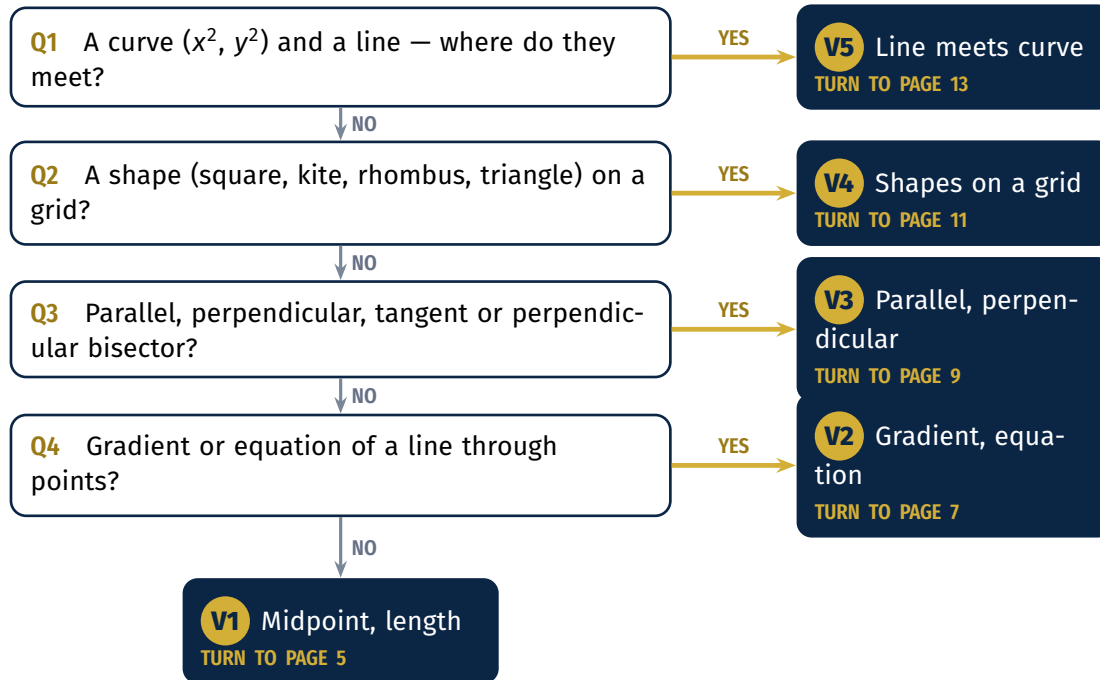
Check a point is on a line by substituting it. Check a gradient by the sketch: line going up left-to-right means $m > 0$.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Midpoint & Length

Midpoint: average. Length: Pythagoras on the differences.

SEEN IN PAST PAPERS

5 / 40

PRACTICE QUESTIONS: 14



Your notes

Came up in: Jan 2020 P1H Q1(a) • Oct/Nov 2020 P1HR Q6 • Jun 2022 P1H Q10 • Nov 2025 P2H Q1

? What is it?**In plain words**

The **midpoint** is halfway: add the x's and halve, add the y's and halve.

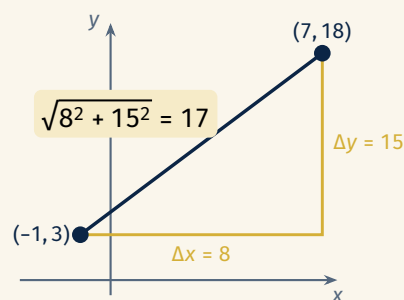
The **length** is the hypotenuse of a right triangle: run Δx , rise Δy . Given the midpoint? Work **backwards**:

$$\frac{11+b}{2} = -19.$$

IT LOOKS LIKE

M of (3, 4) (9, 16)

distance between centres

**How do I do it?****1 Label** (x_1, y_1) and (x_2, y_2) .**2 Midpoint**

Average each coordinate.

3 Length $\Delta x, \Delta y$, then $\sqrt{\Delta x^2 + \Delta y^2}$.**4 Backwards**

Midpoint given? Set each average equal and solve.

AVERAGE • PYTHAGORAS

Squares kill the minus signs.

V1 Midpoint & Length

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“M is the midpoint of PQ. Find M” → **Average the coordinates**

“The midpoint of AB is (a, -19). Find b” → $\frac{y_1 + y_2}{2} = -19$, solve

“Work out the distance between the centres” → **Pythagoras:** $\sqrt{\Delta x^2 + \Delta y^2}$

WORKED EXAMPLE PAST PAPER

June 2022 · Paper 1H · Q10 · 4 marks

Circle C_1 : centre $(-1, 3)$, radius 13 cm. Circle C_2 : centre $(7, 18)$, radius 6 cm. (a) Find the distance between the centres. (b) Explain why the circles intersect.

STEP 1 $\Delta x = 8, \Delta y = 15$

◀ differences

STEP 2 $\sqrt{8^2 + 15^2} = \sqrt{289} = 17$

◀ (a) Pythagoras

STEP 3 $13 - 6 = 7 < 17 < 19 = 13 + 6$

◀ (b) compare with the radii

ANSWER (a) 17 cm (b) $7 < 17 < 19$, so they cross ✓ 8, 15, 17 is a Pythagorean triple ✓

WATCH OUT

× $\Delta y = 18 - 3 = 15, \Delta x = 7 - 1 = 6$

✓ $\Delta x = 7 - (-1) = 8$

Subtracting a negative: $-(-1) = +1$.

EXAM TIP

Two circles cross when the distance between centres is between $R - r$ and $R + r$.

Check yourself

calculator first · answers on page 15

1.1

Nov 2025 P2H

P(3, 4) and Q(9, 16).
Find the midpoint M of PQ.

Answer:

1.2

Oct/Nov 2020 P1HR

A(-3, 11), B(47, b). The
midpoint of AB is
(a, -19). Find a and b.

Answer:

1.3

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A(-3, 5), B(7, -1). Find
the midpoint and the
exact length of AB.

Answer:

V2

VARIANT 2

Gradient & Equation of a Line

Gradient from two points, then

 $y - y_1 = m(x - x_1)$; read m only when y is the subject.

SEEN IN PAST PAPERS

6 / 40

PRACTICE QUESTIONS: 8



Your notes

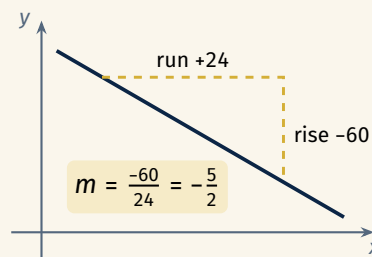
Came up in: Jan 2020 P1H Q1(b)(c) • Jan 2021 P1H Q13(a) • Jan 2022 P1H Q10 • May 2023 P1H Q13

? What is it?

In plain words

Gradient = $\frac{\text{change in } y}{\text{change in } x}$. Then put**one point** and m into $y - y_1 = m(x - x_1)$ and tidy to $y = mx + c$.Given an equation like $2y + 7x = 10$?Make **y the subject** before you read m and c .

IT LOOKS LIKE

 $A(-5, 12)$ $B(19, -48)$ $2y + 7x = 10$ 

How do I do it?

1 Gradient

 $m = \frac{y_2 - y_1}{x_2 - x_1}$ — same order top and bottom.

2 Line

 $y - y_1 = m(x - x_1)$ with one point.

3 Tidy

Expand; $y = mx + c$ or the form asked.

4 Check

The other point must fit.

 $y - y_1 = m(x - x_1)$ One point and m is enough.

V2 Gradient & Equation of a Line

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Find an equation of the line through A and B" → **m first, then $y - y_1 = m(x - x_1)$**

"Line L has equation $2y + 7x = 10$. Find the gradient" → **$y = -\frac{7}{2}x + 5$; $m = -\frac{7}{2}$**

"Does (100, -302) lie on the line?" → **Substitute $x = 100$; compare y**

WORKED EXAMPLE PAST PAPER

May 2023 • Paper 1H • Q13 • 3 marks

A is the point $(-5, 12)$, B is the point $(19, -48)$. Find an equation of the straight line that passes through A and B.

STEP 1 $m = \frac{-48 - 12}{19 - (-5)} = \frac{-60}{24} = -\frac{5}{2}$ ◀ gradient

STEP 2 $y - 12 = -\frac{5}{2}(x + 5)$ ◀ line through A

STEP 3 $y = -\frac{5}{2}x - \frac{25}{2} + 12$ ◀ expand

STEP 4 $y = -\frac{5}{2}x - \frac{1}{2}$ ◀ tidy

ANSWER $y = -\frac{5}{2}x - \frac{1}{2}$

✓ B: $-\frac{95}{2} - \frac{1}{2} = -48$ ✓

WATCH OUT

✗ $2y + 7x = 10$: $m = 7$

✓ $y = -\frac{7}{2}x + 5$: $m = -\frac{7}{2}$

m is the number in front of x only when y is alone.

EXAM TIP

Check the line with the second point — it catches sign slips.

Check yourself

calculator first • answers on page 15

2.1 Jan 2022 P1H

L: $2y + 7x = 10$. Find the gradient and where L crosses the y -axis.

Answer:

2.2 Jan 2020 P1H

Does the point $(100, -302)$ lie on $y = 2 - 3x$? Give a reason.

Answer:

2.3 Jan 2021 P1H

A(5, 8) and B(9, -4). Work out the gradient of AB.

Answer:

V3

VARIANT 3

Parallel & Perpendicular

Parallel: same m . Perpendicular: $-\frac{1}{m}$.
Tangent: perpendicular to the radius.

SEEN IN PAST PAPERS

9 / 40

PRACTICE QUESTIONS: 2


Your notes

Came up in: Jan 2020 P2H Q19 • Jan 2021 P1H Q13(b) • May 2021 P2H Q17 • Jun 2022 P2HR Q20 • Jun 2023 P2HR Q25 • Nov 2023 P1H Q22 • Jun 2025 P2HR Q24 • Nov 2025 P1H Q20

? What is it?

In plain words

Parallel lines have the same gradient. **Perpendicular** lines have gradients that multiply to -1 : flip and change sign.

A **tangent** to a circle is perpendicular to the **radius**. A

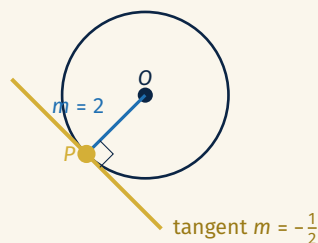
perpendicular bisector goes through the midpoint at $-\frac{1}{m}$.

IT LOOKS LIKE

$$m = 2 \rightarrow -\frac{1}{2}$$

tangent at P

perp. bisector



How do I do it?

1 Gradient you know

From the line or two points.

2 New gradient

Parallel: same. Perpendicular: $-\frac{1}{m}$.

3 Point

Which point is the new line through? (midpoint, P , ...)

4 Line

 $y - y_1 = m(x - x_1)$; give the form asked.**FLIP AND CHANGE SIGN** $m_1 \times m_2 = -1$.

V3 Parallel & Perpendicular

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“**M** is perpendicular to $y = 4x + 7$ ” → $m = -\frac{1}{4}$

“the tangent to the circle at **P**” → **Perpendicular to the radius OP**

“the perpendicular bisector of **PQ**” → **Through the midpoint, gradient $-\frac{1}{m}$**

WORKED EXAMPLE **PAST PAPER**

June 2022 · Paper 2HR · Q20 · 4 marks

The centre **O** of a circle is (4, 7). **A**(6, 11) is on the circle and **AOP** is a diameter. Find an equation of the tangent to the circle at **P**.

STEP 1 $P = (2 \times 4 - 6, 2 \times 7 - 11) = (2, 3)$ ◀ **O** is the midpoint of **AP**

STEP 2 $m_{OP} = \frac{7-3}{4-2} = 2 \Rightarrow m = -\frac{1}{2}$ ◀ tangent $\perp$ radius

STEP 3 $y - 3 = -\frac{1}{2}(x - 2)$ ◀ line through **P**

STEP 4 $y = -\frac{1}{2}x + 4$ ◀ tidy

ANSWER $y = -\frac{1}{2}x + 4$

✓ $2 \times (-\frac{1}{2}) = -1$ ✓

WATCH OUT

× perpendicular to $m = 2$: $m = -2$

✓ $m = -\frac{1}{2}$

Flip as well as change the sign.

EXAM TIP

Midpoint backwards: $P = 2O - A$ (double the centre, take away **A**).

Check yourself

calculator first · answers on page 15

3.1 Nov 2025 P1H

M is perpendicular to $y = 4x + 7$ and passes through (8, 1). Find **M** as $y = mx + c$.

Answer:

3.2 May 2021 P2H

L passes through (4, -1) and (6, 4). **M** $\perp$ **L** meets the y-axis at (0, 8). Where does **M** cross the x-axis?

Answer:

3.3 Jan 2021 P1H

Write down the gradient of a line perpendicular to $y = -4x + 5$.

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM**Shapes on a Grid**

Sketch the shape, list its facts (equal sides, right angles, diagonals), then use V1–V3.

SEEN IN PAST PAPERS

12 / 40

PRACTICE QUESTIONS: 2



Your notes

Came up in: Jan 2020 P1H Q22 • Jan 2020 P2HR Q24 • Oct/Nov 2020 P1H Q22 • Jan 2021 P1HR Q22 • Jan 2021 P2H Q18 • Nov 2021 P1H Q20 • Jan 2022 P2HR Q22 • Jun 2022 P2H Q22 • Jun 2023 P2H Q24 • Jun 2024 P2HR Q21 • Jun 2025 P1H Q25

? What is it?**In plain words**

Each shape gives you

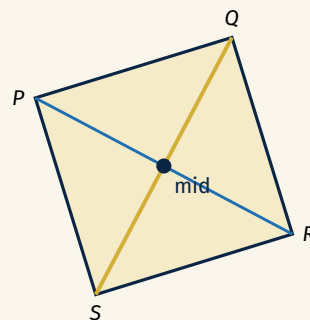
facts about lines: a square has right angles and its diagonals are **perpendicular bisectors** of each other; a kite's diagonals are perpendicular; an isosceles triangle's apex lies on the perpendicular bisector of its base. Turn each fact into a gradient or a midpoint.

IT LOOKS LIKE

square PQRS

isosceles $AB = AC$

rectangle

**QS: PERPENDICULAR BISECTOR OF PR****How do I do it?****1 Sketch**

Rough sketch with all the points.

2 Facts

Equal sides? Right angle? Diagonals?

3 Translate

Right angle $\rightarrow m_1 m_2 = -1$; bisect $\rightarrow$ midpoint; equal $\rightarrow$ length.

4 Calculate

V1–V3; give the form asked.

SHAPE FACT $\rightarrow$ LINE FACT The most common question here.

V4 Shapes on a Grid

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"ABC is isosceles with $AB = AC$; M is the midpoint of BC" → **AM is the perpendicular bisector of BC**

"PQRS is a square; PR is a diagonal" → **$QS \perp PR$ through its midpoint**

"angle ABC = 90° " → **$m_{AB} \times m_{BC} = -1$**

WORKED EXAMPLE PAST PAPER

January 2021 · Paper 1HR · Q22 · 5 marks

ABC is isosceles with $AB = AC$. $B(-1, 5)$, $C(2, 10)$; M is the midpoint of BC. Find an equation of the line through A and M in the form $py + qx = r$.

STEP 1 $M = (0.5, 7.5)$ ◀ midpoint

STEP 2 $m_{BC} = \frac{5}{3} \Rightarrow m_{AM} = -\frac{3}{5}$ ◀ $AM \perp BC$ (isosceles)

STEP 3 $y - 7.5 = -\frac{3}{5}(x - 0.5)$ ◀ line through M

STEP 4 $5y - 37.5 = -3x + 1.5 \Rightarrow 5y + 3x = 39$ ◀ $\times 5$, tidy

ANSWER $5y + 3x = 39$

✓ M: $37.5 + 1.5 = 39$ ✓

WATCH OUT

✗ line AM with gradient $\frac{5}{3}$
✓ $-\frac{3}{5}$

The line from the apex to the midpoint is perpendicular to the base.

EXAM TIP

Integers form: multiply through by the denominators at the very end.

Check yourself

calculator first · answers on page 15

4.1 Jun 2025 P1H

PQRS is a square;
 $P(4, 7)$, $R(8, -5)$. Find
the line through Q and
S as $ay = bx + c$.

Answer:

4.2 Jun 2024 P2HR

ABCD is a square;
 $A(-5, 2)$, $B(3, 5)$. Find
the line through B and
C as $ax + by + c = 0$.

Answer:

4.3 Jan 2021 P2H

Rectangle ABCD:
 $A(-4, -2)$, $B(1, 10)$,
 $C(19, a)$, $D(b, c)$. Find a ,
 b , c .

Answer:

V5

VARIANT 5

Line Meets Curve

Rearrange the line, substitute into the curve, solve the quadratic, then find both points.

SEEN IN PAST PAPERS

8 / 40

PRACTICE QUESTIONS: 2



Your notes

Came up in: Jan 2020 P2H Q22 • Nov 2021 P2H Q19 • Jan 2022 P2HR Q18 • Jun 2023 P2HR Q21 • Nov 2023 P2H Q21 • Jun 2024 P2H Q22 • Nov 2025 P1H Q28

? What is it?**In plain words**

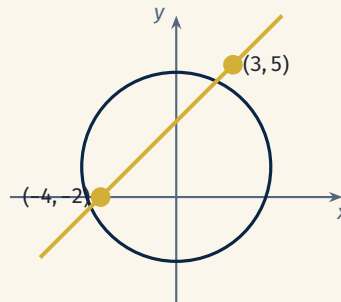
Where they meet,
both equations are true. Make x or y
the subject of the **line**, substitute
into the **curve**: a quadratic.

Two roots → **two points**. Put each
back into the **line** to get the other
coordinate.

IT LOOKS LIKE

$$y = x + 2$$

$$x^2 + y^2 - 2y = 24$$

**How do I do it?****1 Rearrange**Line: $y = \dots$ or $x = \dots$ **2 Substitute**

Into the curve — brackets!

3 SolveExpand, $= 0$, factorise (or formula).**4 Pair up**

Each root back into the line.

LINE INTO CURVE Then pair each x with its y .

V5 Line Meets Curve

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

"The line ... intersects the curve ... at A and B"

→ **Substitute the line into the curve**

"Show clear algebraic working"

→ **Write the quadratic and its factors**

"Find the midpoint of PQ"

→ **Both points first, then average** **WORKED EXAMPLE** **PAST PAPER**

January 2020 · Paper 2H · Q22 · 5 marks

The line $y = x + 2$ intersects the curve $x^2 + y^2 - 2y = 24$ at A and B. Find the coordinates of A and B.

STEP 2 $x^2 + (x + 2)^2 - 2(x + 2) = 24$ ◀ substitute

STEP 3 $2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0$ ◀ expand, tidy

STEP 3 $(x + 4)(x - 3) = 0 : x = 3, x = -4$ ◀ solve

STEP 4 $x = 3 \Rightarrow y = 5; x = -4 \Rightarrow y = -2$ ◀ back into the line

ANSWER (3, 5) and (-4, -2)

$\checkmark 9 + 25 - 10 = 24 \checkmark$

 WATCH OUT

$\times (x + 2)^2 = x^2 + 4$

$\checkmark x^2 + 4x + 4$

Square the whole bracket.

 EXAM TIP

Use the line to find y — the curve can give extra wrong partners.

 Check yourself

calculator first · answers on page 15

5.1 Jun 2024 P2H

$x + y = 5$ meets
 $2x^2 + 3y^2 = 210$. Find
 the points.

Answer:

5.2 Nov 2023 P2H

$x + 2y = 5$ meets
 $x^2 + 3y^2 = 13$ at A and B.
 Find them.

Answer:

5.3 Dr Eslam

Find where $y = x + 1$
 meets $x^2 + y^2 = 25$.

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 $M(6, 10)$

1.2 $a = 22, b = -49$

1.3 $(2, 2); 2\sqrt{34}$

2.1 $-\frac{7}{2}; (0, 5)$

2.2 No: $y = -298$

2.3 -3

3.1 $y = -\frac{1}{4}x + 3$

3.2 $(20, 0)$

3.3 $\frac{1}{4}$

4.1 $3y = x - 3$

4.2 $8x + 3y - 39 = 0$

4.3 $a = 2.5, b = 14,$
 $c = -9.5$

5.1 $(9, -4)$ and $(-3, 8)$

5.2 $(1, 2)$ and $(\frac{23}{7}, \frac{6}{7})$

5.3 $(3, 4)$ and $(-4, -3)$

THE WHOLE TOPIC ON ONE PAGE

Coordinate geometry cheat sheet



Your notes
V1 Midpoint, length 5/40

$$\sqrt{8^2 + 15^2} = 17$$

Average; Pythagoras.

V2 Line 6/40

$$y - y_1 = m(x - x_1)$$

 m from two points; y alone to read m .**V3 Perpendicular 9/40**

$$2 \rightarrow -\frac{1}{2}$$

Flip and change sign; tangent $\perp$ radius.**V4 Shapes 12/40**

$$5y + 3x = 39$$

Shape fact $\rightarrow$ line fact.**V5 Line meets curve 8/40**

$$x^2 + x - 12 = 0$$

Line into curve; pair up.

**TOP 3 LOST MARKS**

1. m read before y is alone $\times$
2. perpendicular: sign changed, not flipped $\times$
3. $(x + 2)^2 = x^2 + 4x$

BOTTOM LINE Sketch and label · average, rise/run, Pythagoras · $-\frac{1}{m}$ · line into curve.